

ECEN 4/5632 Theory and Application of Digital Filters

Lecture 3, 9-01-26

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* With generous help from Gemini Pro and ChatGPT

Continuous-to-Discrete Conversion & Aliasing

How a physical waveform becomes a trustworthy digital array — or permanently loses information.

Where Week 2 lives in the audio → ML pipeline

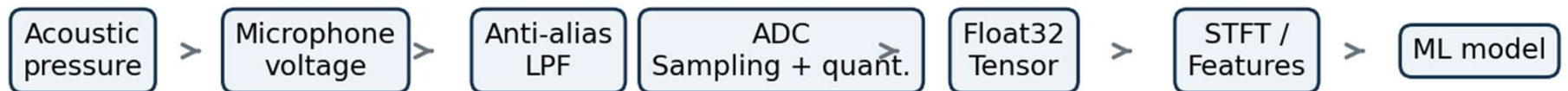


75 minutes • mathematical sampling model • frequency-domain replication • Nyquist • audio/ML aliasing

Where Week 2 sits in the audio → ML pipeline

Everything downstream assumes the ADC delivered faithful samples.

Where Week 2 lives in the audio → ML pipeline



Aliasing is especially dangerous for ML because it can create repeatable, structured features that are physically false.

Learning outcomes

By the end of Lecture 3, students should be able to derive—not just quote—the sampling theorem.

- Model ideal sampling as multiplication by a periodic impulse train.
- Derive the replicated spectrum $X_s(j\Omega)$ from the CTFT of the impulse train.
- State the no-overlap condition and explain why it enables perfect reconstruction.
- Predict the alias frequency of a sinusoid at a given sample rate.
- Explain why anti-alias filtering must occur before sampling or decimation.

The C/D interface: from volts to an indexed sequence

A microphone produces a continuous-time voltage; software sees numbers.

discrete-time (DT) sequence

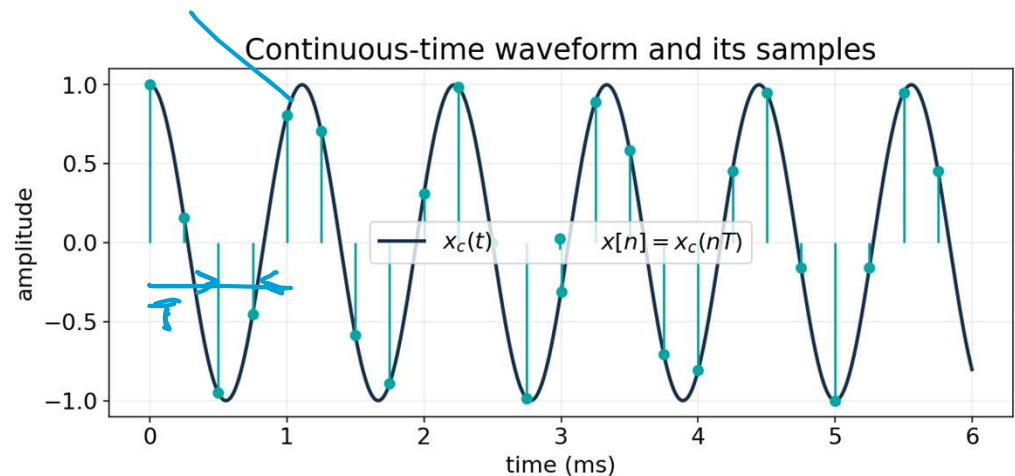
$$\underline{x[n] = x_c(nT)}, \quad T = 1/f_s$$

- n is an integer sample index.
- T has units of seconds/sample.
- $f_s = 1/T$ has units of samples/second (Hz).
- Sampling discretizes time; quantization will discretize amplitude.

n corresponds to different t depending on $f_s = \frac{1}{T}$

sampling rate

CT $x_c(t)$ continuous-time waveform



Continuous frequency and discrete frequency are related—but not identical

Sampling converts rad/s to rad/sample.

Ω ω frequency rad/sec

$$e^{j2\pi n} = 1$$

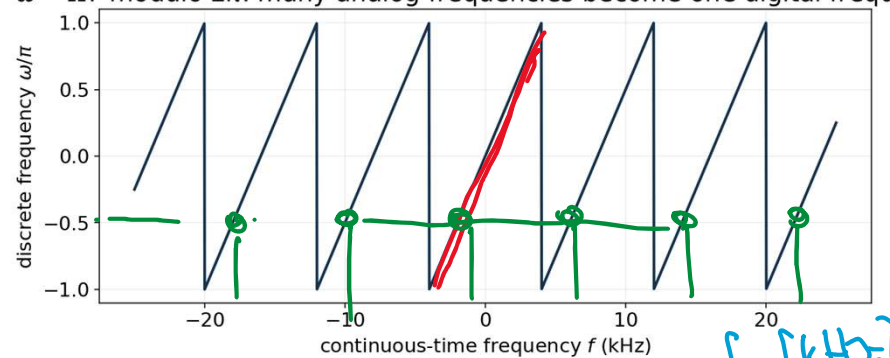
2π freq.

$$\omega = \Omega T = 2\pi \frac{f}{f_s}$$

- Ω : continuous-time angular frequency [rad/s].
- ω : discrete-time angular frequency [rad/sample].
- ω is only unique modulo 2π .
- Therefore many analog frequencies can map to one digital frequency.

$$\frac{\omega}{2\pi} = \frac{f}{f_s}$$

$\omega = \Omega T$ modulo 2π : many analog frequencies become one digital frequency



f [kHz]

ω : rad/sample $\omega = 2\pi \frac{f}{f_s}$
 f : frequency in Hz cycles/sec

f cycles/sec
 f_s samples/sec

$\frac{f}{f_s}$: normalized frequency cycles/sample

Ideal impulse-train sampling

The mathematical sampler is a multiplier.

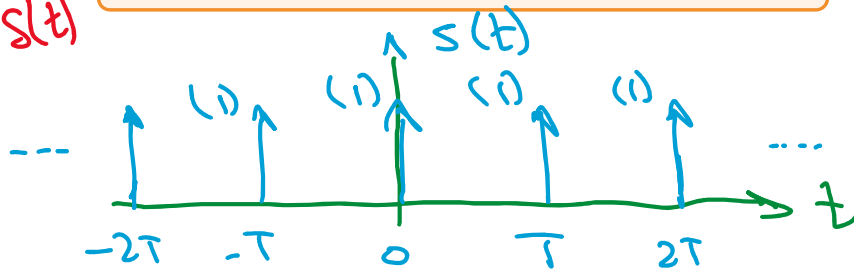
$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$x_s(t) = x_c(t)s(t)$$

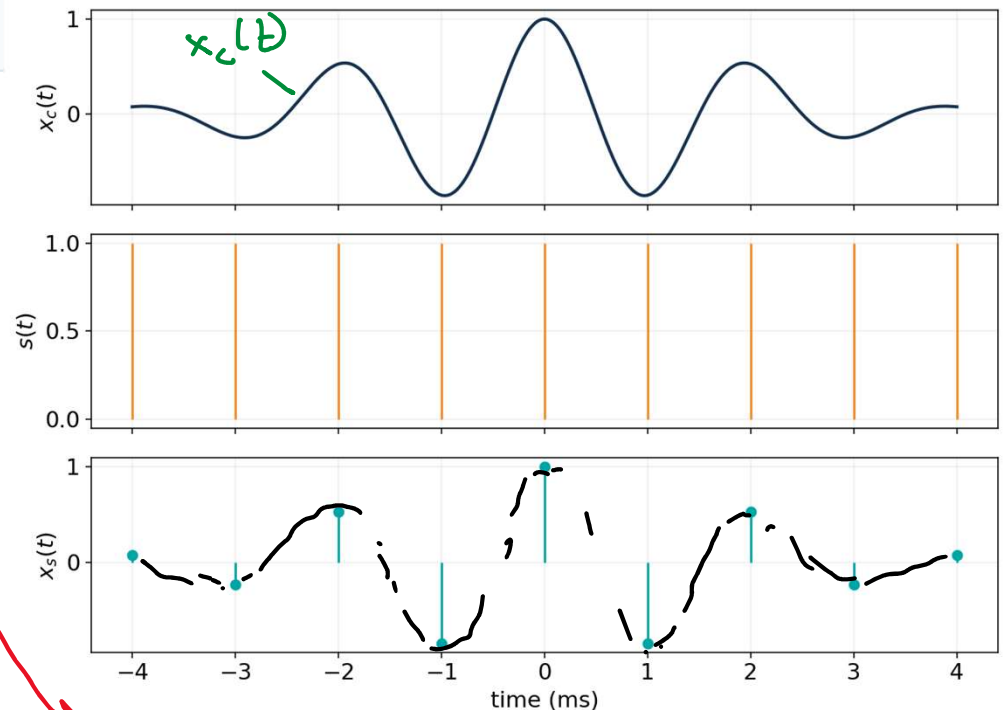
$$\begin{aligned} x_s(t) &= x_c(t) \cdot \sum \delta(t - nT) \\ &= \sum x_c(t) \delta(t - nT) \\ &= \sum x_c(nT) \cdot \delta(t - nT) \end{aligned}$$

sifting property of $\delta(t)$

The impulse train is an analysis device—not a literal ADC circuit.



Ideal impulse-train sampling: multiply in time



$$x[n] = x(nT) \Rightarrow \underline{x_s(t) = \sum x[n] \cdot \delta(t - nT)}$$

Why impulses? The sifting property does the bookkeeping for us

Multiplying by $\delta(t-nT)$ keeps only the value at $t=nT$.

$$\underline{x_c(t)} \delta(t - nT) = x_c(nT) \delta(t - nT)$$

$$\underline{x_s(t)} = \sum_{n=-\infty}^{\infty} x_c(nT) \delta(t - nT) = \sum_{\underline{n}} x[n] \delta(t - nT)$$

Each impulse location records one sample value. The sequence $x[n]$ is encoded in the impulse weights.

Example: $x_c(t) = \cos(2\pi 1000 t)$ $f_s = 8000$ samples/sec

$$x_c(nT) = \cos\left(2\pi \frac{1000 n}{8000}\right) = \cos\left(\frac{\pi n}{4}\right) = x[n]$$

$$x_s(t) = \sum_n \cos\left(\frac{\pi n}{4}\right) \delta(t - nT)$$

From Oppenheim/Schafer Book

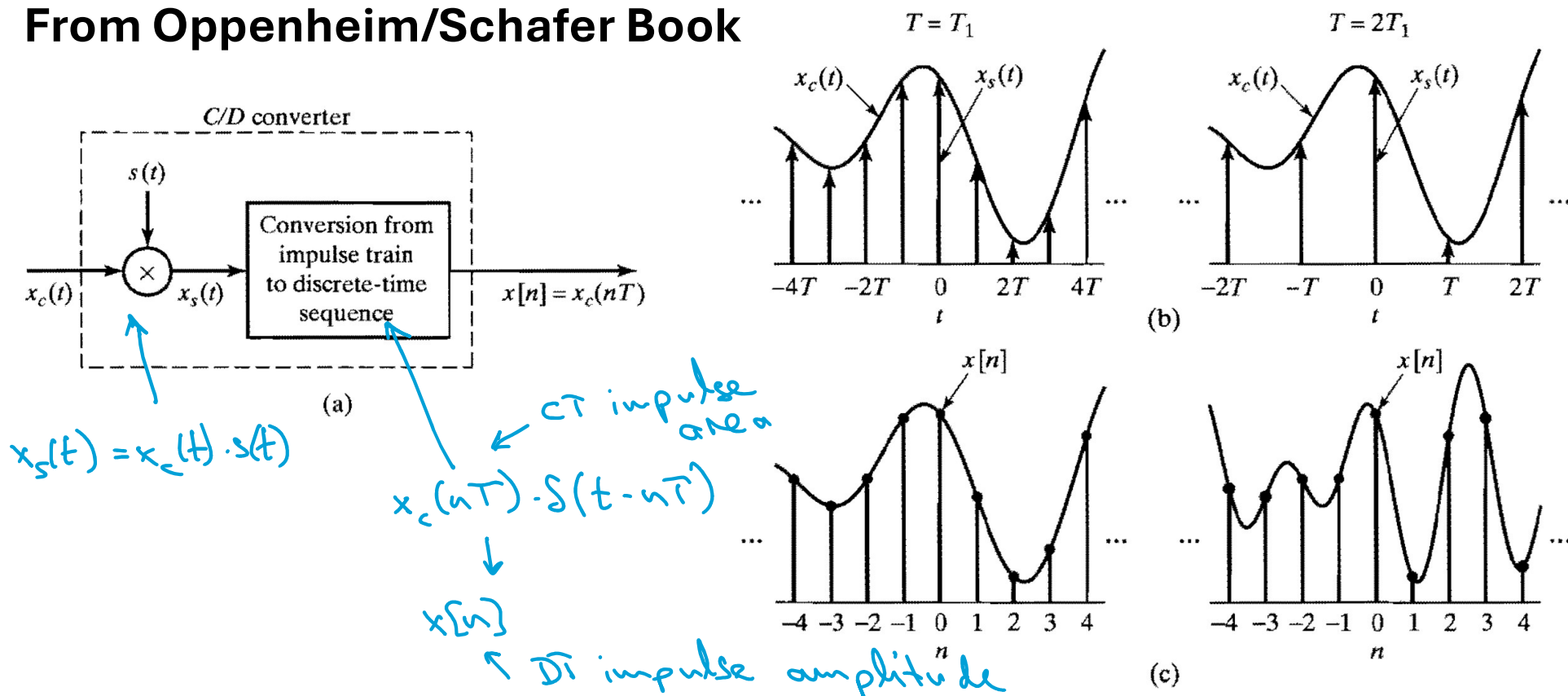
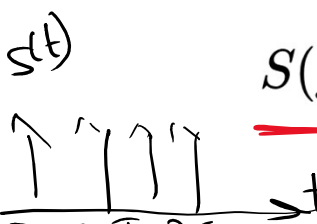


Figure 4.2 Sampling with a periodic impulse train, followed by conversion to a discrete-time sequence. (a) Overall system. (b) $x_s(t)$ for two sampling rates. (c) The output sequence for the two different sampling rates.

The impulse train is periodic in both time and frequency

A periodic train of impulses transforms to another impulse train.

$$\Omega_s = \frac{2\pi}{T}$$



$$s(t) = \sum_n \delta(t - nT)$$
 periodic period T

$$S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

$$\text{IFS} \rightarrow = \sum_k S[k] e^{j2\pi kt/T}$$

$$S(j\Omega) = \mathcal{F} \left\{ \sum_k S[k] e^{j2\pi kt/T} \right\}$$

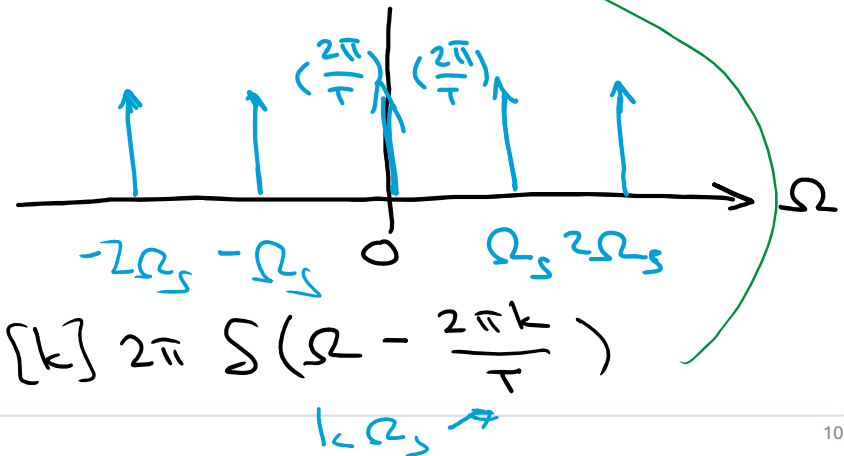
$$= \sum_k S[k] \mathcal{F} \left\{ e^{j2\pi kt/T} \right\} = \sum_k S[k] 2\pi \delta\left(\Omega - \frac{2\pi k}{T}\right)$$

$$= \frac{2\pi}{T} \sum_k \delta(\Omega - k\Omega_s)$$

Time spacing $T \leftrightarrow$ frequency spacing $\Omega_s = 2\pi/T$

$$S[k] = \frac{1}{T} \int_T s(t) e^{-j2\pi kt/T} dt$$

This is the mechanism that creates spectral replicas.



Sampling in time becomes convolution in frequency

Now the central derivation is one line of CTFT algebra.

$$x_s(t) = x_c(t)s(t)$$

← multiplication
by train of impulses
in time

$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega)$$

Sampling creates shifted copies of $X_c(j\Omega)$, spaced by Ω_s and scaled by $1/T$.

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

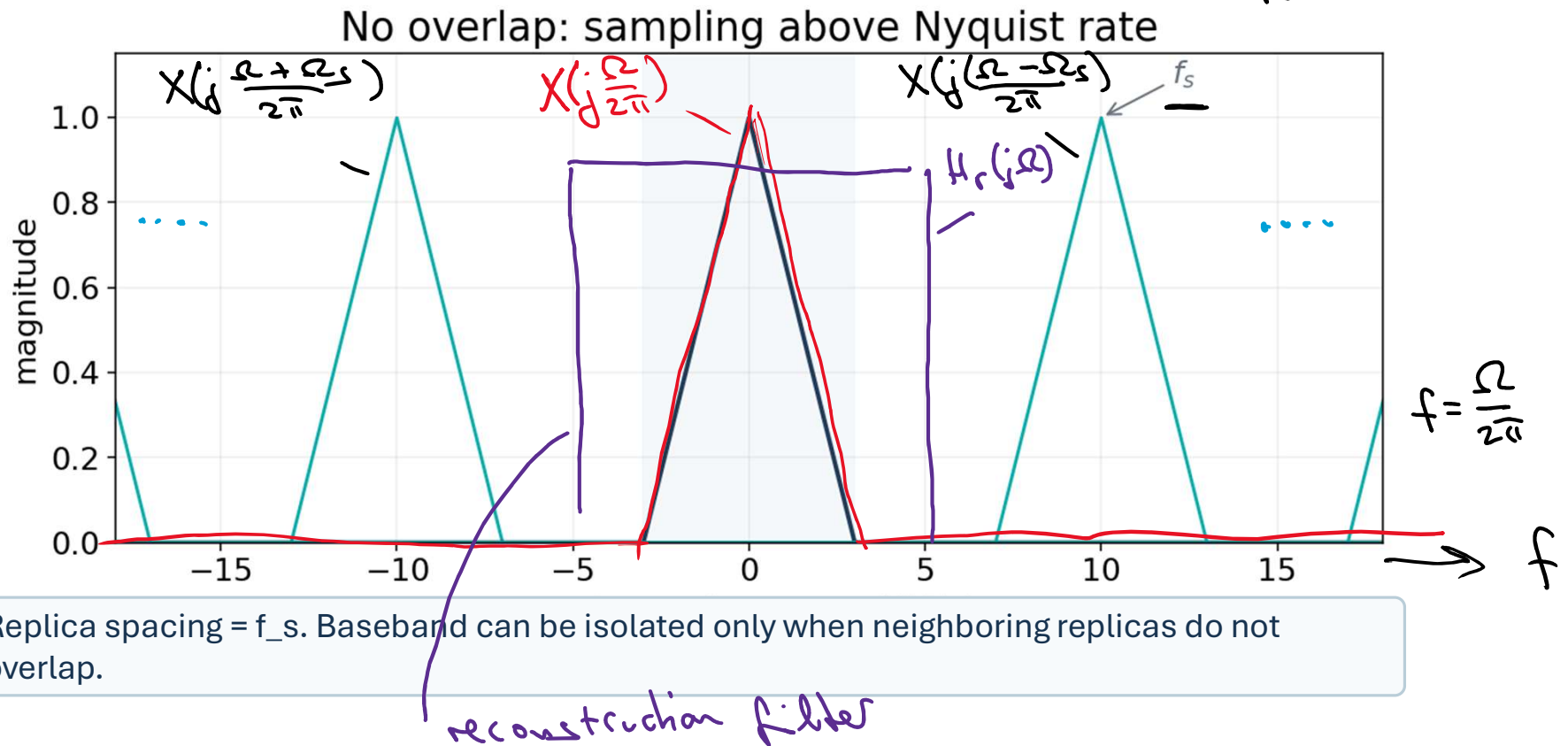
$X_c(j\Omega)$
shifted left and right
by $k\Omega_s$

convolution by train of
impulses in frequency

What the sampled spectrum looks like

Every copy contains the same analog information—until the copies overlap.

$$f_s = 10 \text{ kHz}$$



Nyquist-Shannon: the no-overlap condition

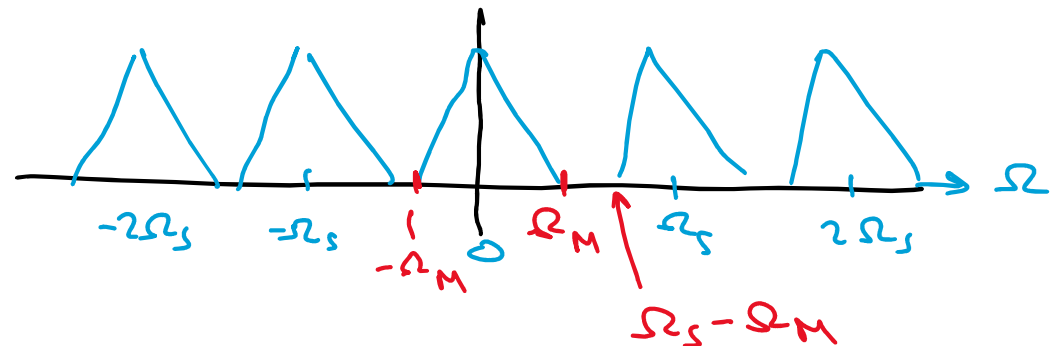
Let Ω_M be the largest analog frequency with nonzero spectrum.

$$\Omega_s > 2\Omega_M$$

$$f_s > 2f_M$$

- If the condition holds, one spectral replica can be isolated by an ideal low-pass filter.
- At exact equality, idealized edge cases require care; practical systems use transition bands and margin.
- The “Nyquist frequency” $f_s/2$ is a property of the sample rate; the “Nyquist rate” $2f_M$ is a property of the signal bandwidth.

For no overlap
 $\Omega_s - \Omega_M > \Omega_M$
 $\Rightarrow \underline{\Omega_s > 2\Omega_M}$

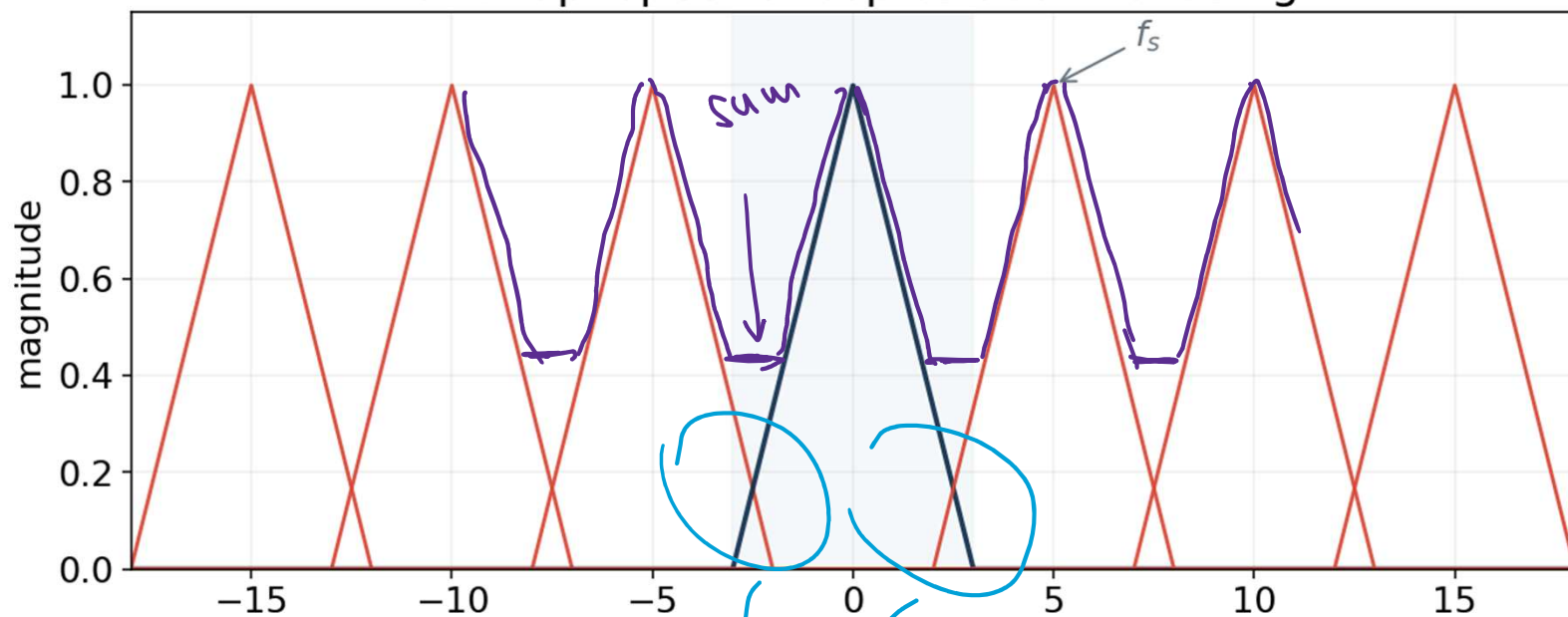


Aliasing: neighboring replicas overlap and add

After overlap, there is no unique way to separate the original copies.

$$X_s(j\Omega) = \sum_k X_c(j(\Omega - k\Omega_s))$$

Overlap: spectral copies add → aliasing



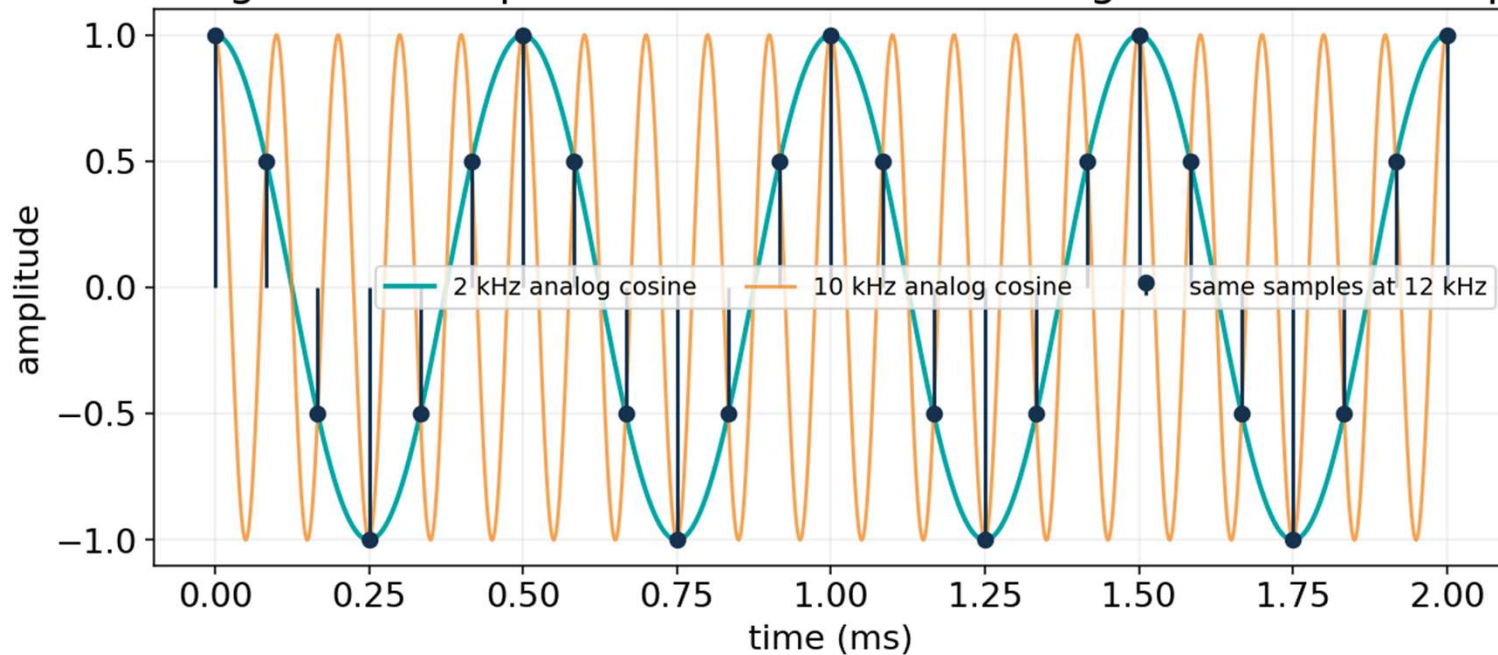
This is irreversible at the sample level: multiple continuous-time spectra are consistent with the same samples.

Aliasing is non-uniqueness—not ordinary additive noise

Two different analog sinusoids can produce exactly the same sequence.

$$f_s \approx 12 \text{ kHz}$$

Aliasing is non-uniqueness: 2 kHz and 10 kHz give identical samples



$$e^{j(\omega+2\pi m)n} = e^{j\omega n}$$

Worked audio example: 10 kHz sampled at 12 kHz

New Nyquist frequency is only 6 kHz.

$$f_{alias} = |f_0 - kf_s| \quad \text{fold into } [0, f_s/2]$$

- The samples contain a 2 kHz discrete-time cosine.
- There is no label attached saying “this used to be 10 kHz.”
- A spectrogram will confidently display energy at 2 kHz.

$$10 \text{ kHz} - 12 \text{ kHz} = -2 \text{ kHz} \Rightarrow 2 \text{ kHz}$$

$$f_s = 12 \text{ kHz}$$

Prediction before demo: what will happen to 7 kHz? 9 kHz? 13 kHz?

f_0	f_{alias}
10k	$ 10 - 12 = 2 \text{ k}$
5k	$ 5 - 0 = 5 \text{ k}$
7k	$ 7 - 12 = 5 \text{ k}$

The same fact in normalized frequency

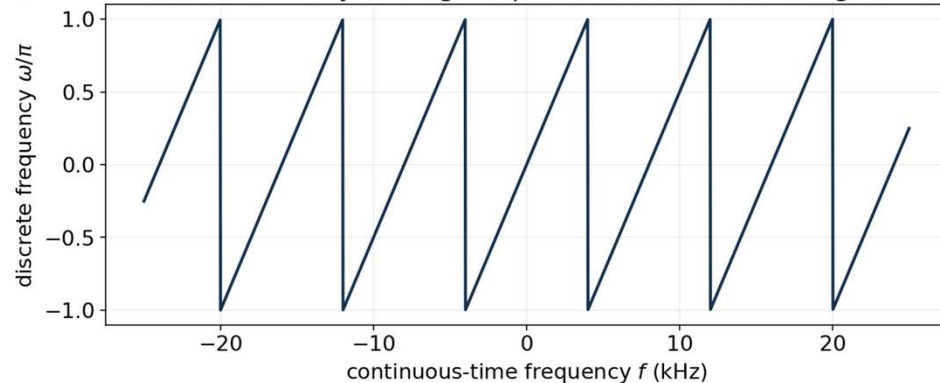
The mapping $\omega = \Omega T$ is periodic modulo 2π .

$$\omega = 2\pi \frac{f}{f_s} \pmod{2\pi}$$

At $f_s = 12$ kHz:
 10 kHz $\rightarrow 5\pi/3 \equiv -\pi/3 \rightarrow$ real-cosine alias
 at $+\pi/3 \rightarrow 2$ kHz.

$$f_s = 12k$$

$\omega = \Omega T$ modulo 2π : many analog frequencies become one digital frequency



$$10 \text{ kHz} \rightarrow \frac{f}{f_s} = \frac{10000}{12000} = \frac{5}{6}$$

$$\omega = 2\pi \cdot \frac{5}{6} = \frac{5\pi}{3} \underset{\substack{\uparrow \\ \text{modulo } 2\pi}}{=} \frac{5\pi}{3} - 2\pi = \frac{5\pi - 6\pi}{3} = -\frac{\pi}{3}$$

$$\cos\left(\frac{5\pi}{3}\right) = \cos\left(-\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right)$$

cos is even function

Python Demo

```
f00, fs = 900.0, 4000.0

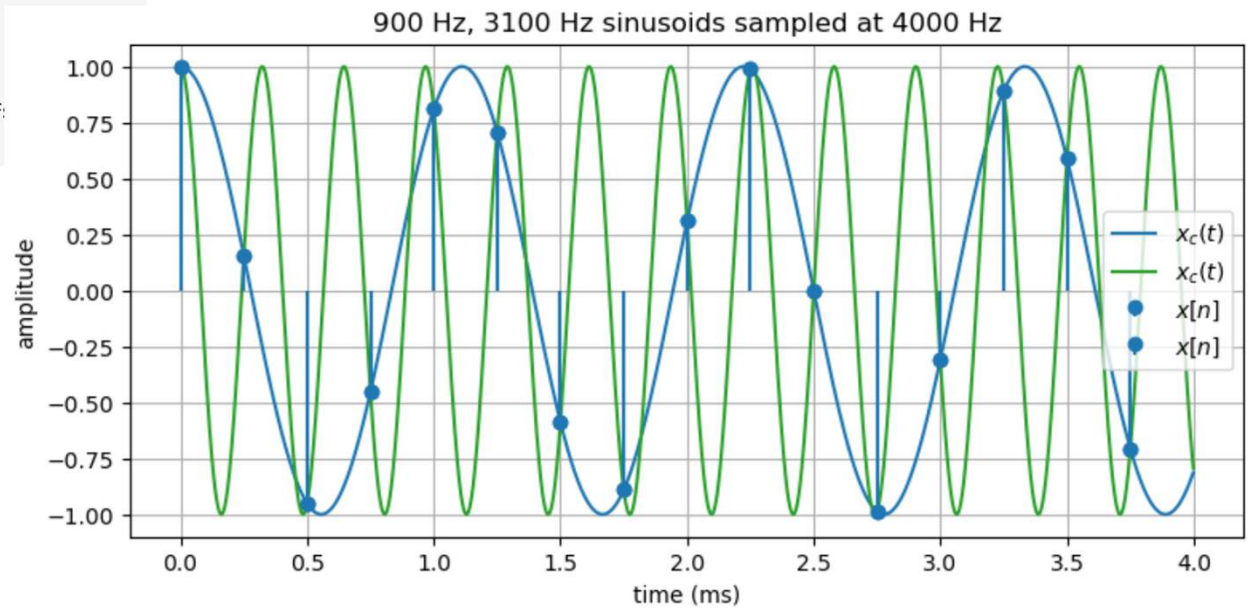
t, xc, n, tn, xn = sample_sinusoid(f00, fs, duration_s=0.004)

plt.plot(t * 1000, xc, label=r"$x_c(t)$")
plt.stem(tn * 1000, xn, basefmt=" ", label=r"$x[n]$")

f01, fs = 3100.0, 4000.0
t, xc, n, tn, xn = sample_sinusoid(f01, fs, duration_s=0.004)

plt.plot(t * 1000, xc, label=r"$x_c(t)$")
plt.stem(tn * 1000, xn, basefmt=" ", label=r"$x[n]$")

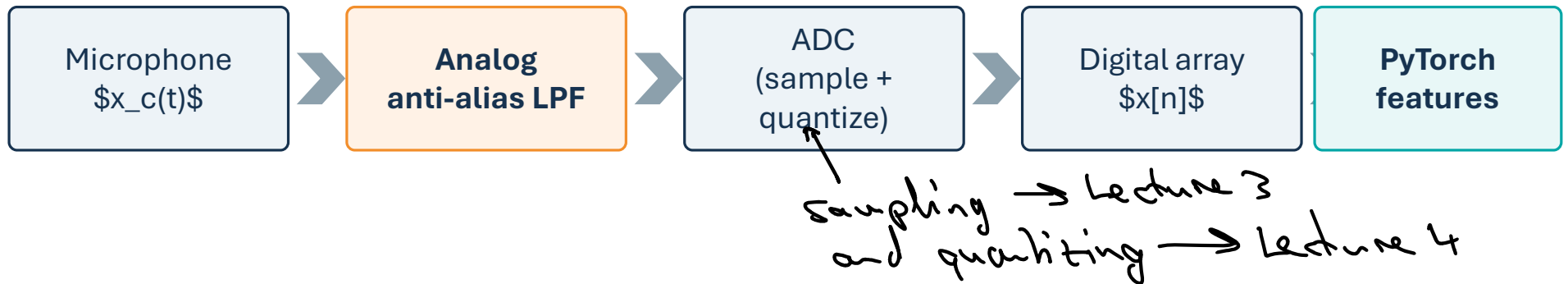
plt.xlabel("time (ms)"); plt.ylabel("amplitude")
plt.title(f"{f00:.0f} Hz, {f01:.0f} Hz sinusoids sampled at {f00:.0f} Hz")
plt.legend(); plt.show()
```



Digital frequency: $\omega_{00} = 1.414$ rad/sample = 0.450 π rad/sample
Digital frequency: $\omega_{01} = 4.869$ rad/sample = 1.550 π rad/sample

The anti-alias filter belongs before the sampler

Once energy above $f_s/2$ has folded, digital filtering cannot know its origin.



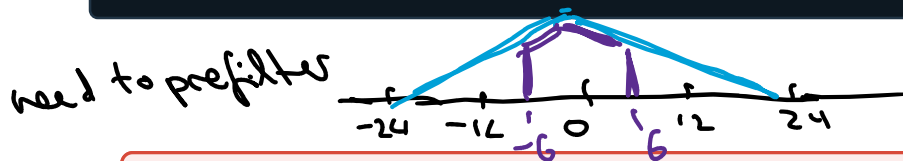
Filter cutoff is designed below $f_s/2$ so the real analog transition band fits before Nyquist.

Downsampling is another sampling problem

Going from 48 kHz $\rightarrow$ 12 kHz creates a new Nyquist limit at 6 kHz.

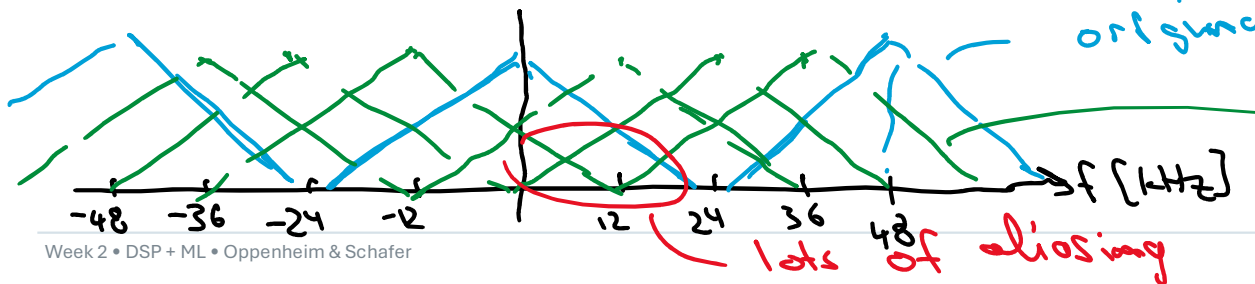
Naive: keep every 4th sample

```
x_12k = x_48k[::4]
# fast... but aliases > 6 kHz
```



A 10 kHz component becomes 2 kHz after naive 4 $\times$ decimation.

$$f_{s1} = 48 \text{ kHz} \rightarrow f_{s2} = 12 \text{ kHz} = \frac{f_{s1}}{4}$$



Correct: low-pass, then keep every 4th

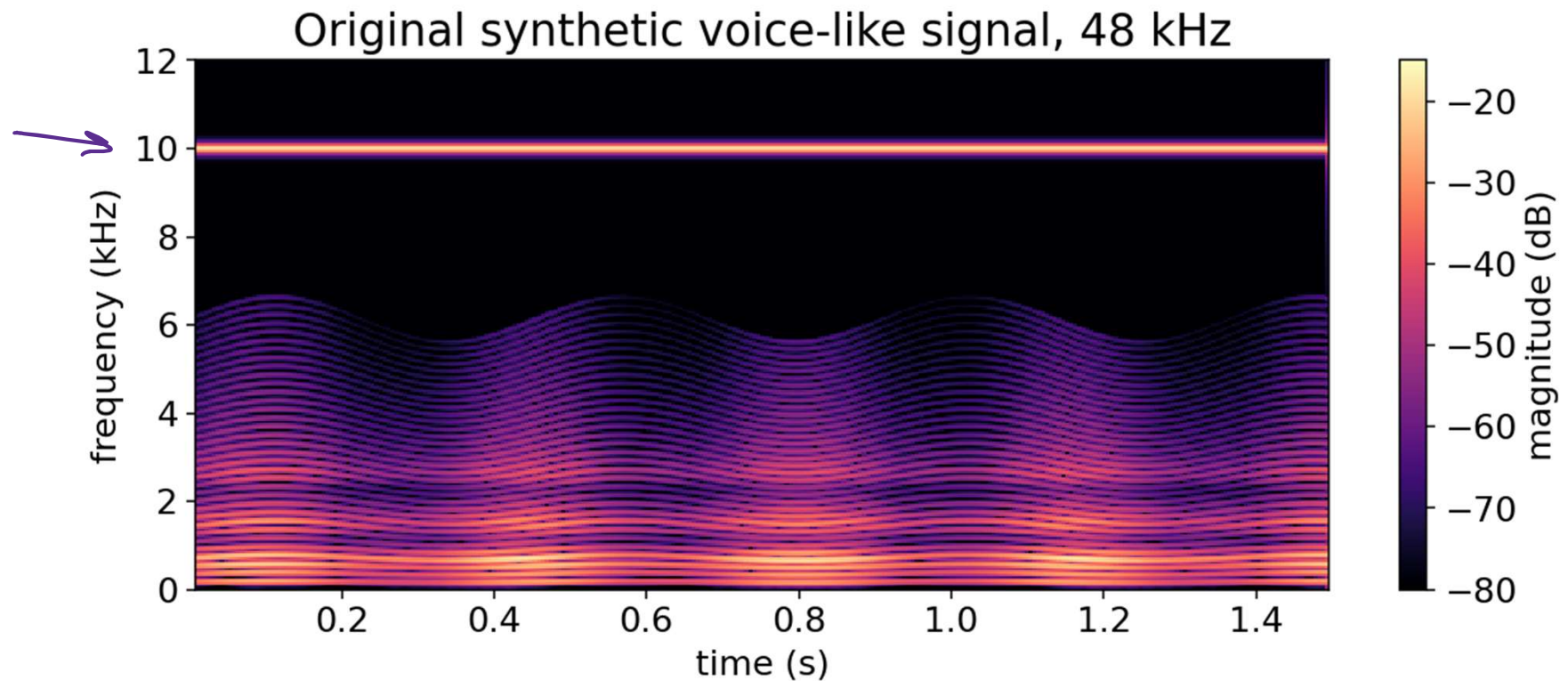
```
x_12k = signal.resample_poly(
    x_48k, up=1, down=4)
# FIR anti-alias filter + decimation
```

The anti-alias filter removes energy above the new 6 kHz Nyquist frequency before decimation.

original $X_s(j\Omega)$ @ $f_s = 48 \text{ kHz}$
added when we go
from f_{s1} to f_{s2} w/o
filtering

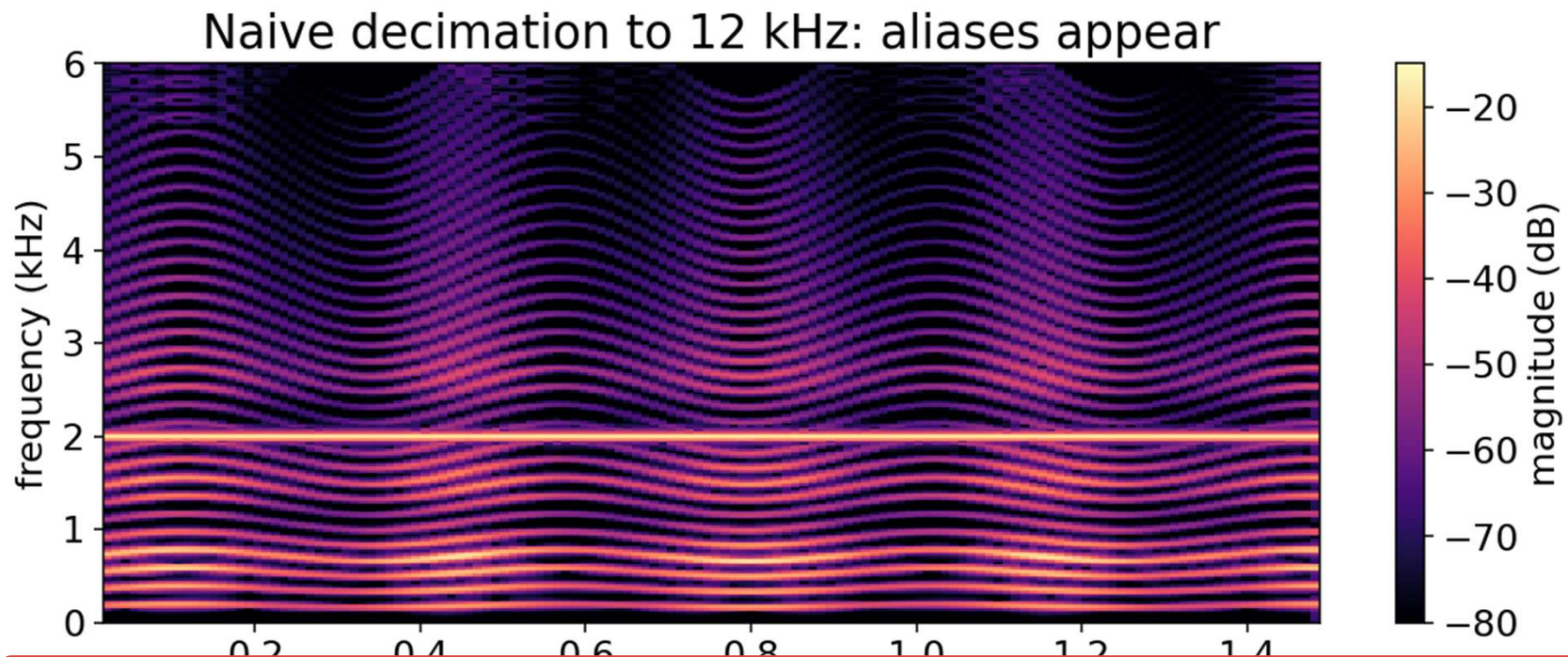
What the ML system “sees”: original 48 kHz spectrogram

The synthetic voice-like signal contains harmonics plus a deliberate 10 kHz component.



Naive 4× decimation creates false low-frequency structure

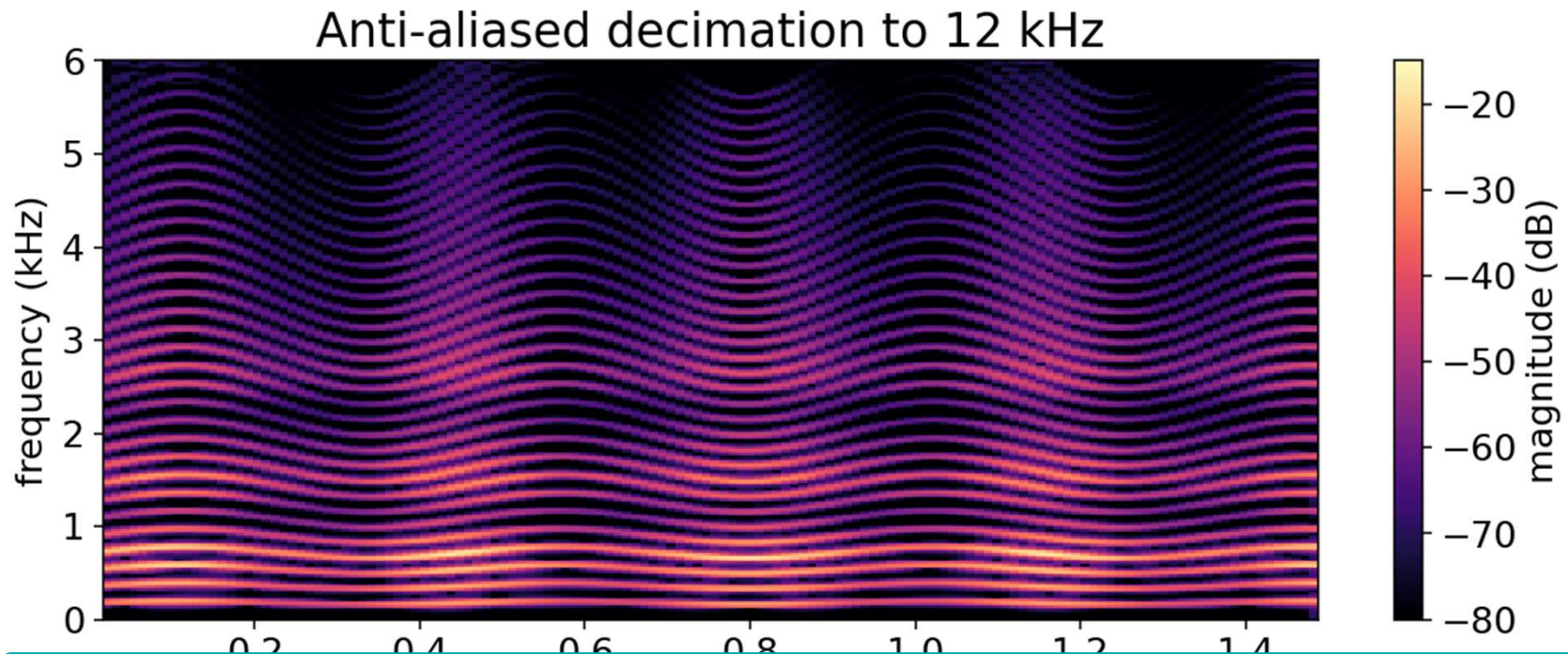
The 10 kHz component folds to 2 kHz at the new 12 kHz sample rate.



A CNN has no way to know this bright 2 kHz structure is an alias rather than a real acoustic feature.

Anti-aliased decimation preserves a physically meaningful spectrum

Filter first, then change the sampling grid.



We lose content above 6 kHz by design—but we do not mislabel that content as lower-frequency energy.

Think–Pair–Share: three checks before data enters a model

2 minutes individually • 2 minutes in pairs • quick debrief

1 A sensor signal is bandlimited to 18 kHz. Is $f_s = 32$ kHz sufficient for ideal sampling?

2 A 17 kHz tone is sampled at 20 kHz. What alias frequency appears in $[0, 10]$ kHz?

3 Why can a digital low-pass filter after the ADC not “remove aliasing” in general?

Lecture 3 summary: sampling is an information-preservation theorem

The mathematical model explains exactly when the digital sequence remains invertible.

- Ideal sampling: multiply $x_c(t)$ by an impulse train.
- Frequency domain: $X_c(j\Omega)$ is replicated every Ω_s and scaled by $1/T$.
- No overlap requires $f_s > 2f_M$ for a bandlimited signal.
- Aliasing means different analog signals produce the same samples—irreversible without extra assumptions.
- Anti-alias filtering must happen before sampling or rate reduction.

Next: If the samples are valid, how do we turn them back into a continuous waveform—and what does finite bit depth do?